

DRAFT — DBA-OIL-FC4: Crude and Product Transport

Crude and Product Transport

Trunk and Product Pipelines

Design Basis Accident — Military Action

DBA-OIL-FC4 — Facility Class Document

Issued Under: DBA-OIL Sector Framework

Version 0.1 — DRAFT

June 2026

Timothy E. Roxey

Eclectic Technologies

Prepared for: Petroleum Sector — Industry, Government, and Allied Partners

Parent document: DBA-OIL Sector Framework v1.5

Preface

The petroleum pipeline is the invisible artery of the American economy. It moves crude oil from wellheads to refineries and refined products from refineries to the terminals where consumers ultimately draw supply. It does this continuously, underground, across thousands of miles, without the visibility of a refinery stack or an offshore platform. When it works, no one notices. When it stops, everyone does — within days.

The Colonial Pipeline ransomware attack of May 2021 established a principle that this document takes as its analytical foundation: a sophisticated adversary does not need to physically destroy a pipeline to shut it down. An operator who cannot trust its own control systems will shut down voluntarily, because the alternative — running a hazardous liquid pipeline blind — is operationally and legally untenable. The adversary who understands this dynamic has a consequence-equivalent attack vector that requires no explosive, no kinetic action, and no physical access to the right-of-way. The six-day Colonial shutdown, driven entirely by an IT-side ransomware event with no damage to the pipeline itself, produced regional fuel shortages across the Eastern Seaboard. A state actor with the same objective and no constraint on kinetic action can produce the same supply disruption with a physical repair timeline measured in months rather than days.

This document establishes the Design Basis Accident scenarios for deliberate military action against crude and product trunk pipeline infrastructure. It addresses three analytically distinct scenarios: the cyber-kinetic product pipeline attack that combines SCADA compromise with physical pump station destruction to prevent restart; the multi-point kinetic rupture of a major crude trunk pipeline at locations chosen to exceed the pipeline operator's simultaneous repair capacity; and the compound attack on intersecting crude and product pipeline corridors, where the simultaneous loss of both systems eliminates the supply chain redundancy that each provides for the other. Overlying all three is the global supply context: a Hormuz closure or Persian Gulf supply disruption that degrades the import alternatives and emergency response mechanisms that would otherwise limit the domestic consequence of any single pipeline attack.

The pipeline is the one facility class in the DBA-OIL series where the adversary's most powerful weapon is not a munition. It is a keyboard — and the Design Basis must account for both.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the crude and product transport facility class. Its purpose is to define the bounding threat event that pipeline security planning, resilience investments, and recovery architecture must address. As

with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A pipeline security program designed against a single-vector physical threat — right-of-way intrusion, pipeline rupture at an accessible crossing — will fail when a sophisticated adversary combines SCADA pre-positioning with precision kinetic action against pump stations and simultaneous attacks at multiple pipeline segments chosen to exceed the operator's repair capacity and exhaust the emergency response system's resources. The DBA-MA approach defines the coordinated, multi-vector attack exploiting both cyber and physical vectors as the governing design event.

1.2 Scope and Applicability

This document addresses crude and product trunk pipeline systems: the large-diameter, high-pressure liquid pipeline networks that move crude oil from domestic producing regions and import terminals to refineries, and move refined petroleum products from refineries to distribution terminals. The facility class encompasses mainline pump stations, pipeline segments including water and environmentally sensitive crossings, SCADA and pipeline control systems, and the associated measurement and regulation infrastructure that governs product custody and flow management.

Offshore export pipelines from production platforms to the landfall point are addressed in DBA-OIL-FC2. Terminals and storage facilities at pipeline origin and terminus points are addressed in DBA-OIL-FC3. Natural gas transmission pipelines and compressor stations are addressed in DBA-GAS-FC1; there is no overlap in the commodity stream, but the pipeline rupture and compressor station loss scenarios in DBA-GAS-FC1 provide an analytical parallel for the non-adversarial bounding events in this document. Where FC4 pipeline events produce consequences that propagate to connected terminals or refineries, this document addresses those consequences as they originate at the pipeline without displacing the dedicated facility-class analysis for the receiving facility.

1.3 Relationship to Governing Documents

This document is issued under the DBA-OIL Sector Framework (v1.5), which is itself issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the pipeline-specific physical configuration, asset taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

The inheritance map for this document is as follows. The threat actor taxonomy (Tier 1, 2, 3), the four characteristics of military action (intentionality, persistence, precision, denial), the three-tier

consequence framework, and the adequacy standards are inherited unchanged from the sector framework. The bounding event categories are inherited with pipeline-specific adaptation. The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the pipeline SCADA environment in Section 3.3. The TSA mandatory cybersecurity directives for pipeline operators, issued following the Colonial Pipeline event, provide the regulatory foundation on which the physical security design basis in Section 3.2 is built.

1.4 The Pipeline as the Invisible Artery — The Voluntary Shutdown Problem

A fundamental characteristic of the crude and product pipeline facility class distinguishes it from every other facility class in the DBA-OIL series: the adversary's most effective attack vector may not require physical contact with the pipeline at all. The Colonial Pipeline event demonstrated that SCADA or IT compromise — even compromise that does not directly affect pipeline control systems — can produce a voluntary operator shutdown whose supply consequence is indistinguishable from a physical rupture. The operator's decision calculus is straightforward: a pipeline operator who cannot trust the integrity of its control system data cannot safely manage product flow, interface points, or custody transfer. Shutdown is the only responsible option.

This creates an attack surface that is qualitatively different from the attack surface of any other petroleum facility class. A refinery or terminal can be partially operated under manual procedures when automation systems fail. A large-diameter hazardous liquid pipeline operating at high pressure across thousands of miles of right-of-way cannot be safely managed manually at scale. The pipeline's operational dependency on its control system is structural, not procedural — and the adversary who compromises that control system achieves a supply disruption consequence without incurring the detection risk, physical access requirement, or attribution exposure of a kinetic attack.

The DBA-MA bounding event must therefore account for two distinct attack pathways: the cyber-only pathway that exploits the voluntary shutdown dynamic, and the cyber-kinetic pathway that adds physical pump station destruction to extend the outage beyond the SCADA reconstitution timeline. The second is the DBA-MA bounding event; the first is the threshold below which the non-adversarial Design Basis applies and above which the MA extension begins.

2. Pipeline Profile and Asset Taxonomy

2.1 Pipeline Types

Crude trunk pipelines. Large-diameter, high-pressure pipelines that move crude oil from domestic producing regions — the Permian Basin, Bakken formation, Gulf of Mexico offshore landfall points — to Gulf Coast and Midwest refineries. Major crude trunk systems include the

Enbridge Mainline (the largest crude pipeline system in North America, moving Canadian and Bakken crude to Midwest and Gulf Coast refineries), the Dakota Access Pipeline (Bakken crude to Gulf Coast markets), and the Seaway and Longhorn systems. Crude trunk pipelines typically operate at pressures of 500 to 1,200 psi with diameters of 24 to 48 inches. A single major crude trunk line can move 500,000 to 1,000,000 barrels per day; its loss eliminates refinery feedstock supply that cannot be quickly replaced through alternative routing.

Product pipelines. Pipelines carrying refined petroleum products — gasoline, diesel, jet fuel, heating oil — from refinery terminals to distribution points. Colonial Pipeline (Line 1: gasoline; Line 2: distillates) is the largest and most consequential product pipeline system in the United States, carrying approximately 100 million gallons per day along a 5,500-mile corridor from Houston, Texas to Linden, New Jersey. The Plantation Pipeline parallels portions of Colonial in the Southeast. Product pipelines move multiple products in batches separated by interface zones; SCADA management of batch tracking and custody transfer is operationally critical and cannot be performed manually across a system of Colonial's scale.

Gathering pipelines. Smaller-diameter pipelines that aggregate production from individual wells and deliver it to trunk pipeline injection points or processing facilities. Gathering pipeline disruption affects production from individual fields but does not produce the systemic supply consequences of trunk pipeline loss. Gathering pipelines are addressed as upstream infrastructure in DBA-OIL-FC2 for offshore gathering and are outside the primary scope of this document.

2.2 Pump Stations

Pump stations are the operational nodes of the liquid pipeline system. They maintain the pressure differential that drives product flow through the pipeline and are the primary control point for flow rate management. A major product pipeline like Colonial Pipeline operates approximately 40 pump stations along its 5,500-mile route; each station contains electric motor-driven or turbine-driven centrifugal pumps, associated valves and piping, electrical switchgear, and local control systems connected to the pipeline SCADA network.

Pump stations are the highest-consequence physical attack target on a liquid pipeline because they are the nodes that, if destroyed, cannot be bypassed. A pipeline rupture at a remote right-of-way location can sometimes be isolated and repaired while adjacent pipeline segments continue to operate. A pump station destroyed at a critical chokepoint — particularly at a terrain-constrained location where the pipeline crosses a mountain range, a major river, or an elevation change that requires pumping — stops all flow through that segment regardless of the physical integrity of the pipe on either side. The adversary who understands pipeline hydraulics selects pump stations at hydraulic chokepoints as primary targets.

2.3 SCADA and Pipeline Control Architecture

Liquid pipeline SCADA systems monitor pressure, flow rate, temperature, and product quality at pump stations, block valve stations, and measurement points along the right-of-way. They manage pump start and stop commands, valve actuations, and batch tracking across the full pipeline length from a centralized control room. Colonial Pipeline's control center in Alpharetta, Georgia manages the entire 5,500-mile system from a single facility; loss of that control center — through cyber compromise, physical attack, or communications disruption — affects the entire pipeline simultaneously.

The Colonial Pipeline ransomware event demonstrated the architecture's vulnerability at the IT-OT boundary: the ransomware infected billing and IT systems, not the pipeline control system itself, but the operator's inability to trust the integrity of the broader network environment forced a precautionary shutdown of the pipeline SCADA system. TSA's subsequent mandatory cybersecurity directives require pipeline operators to implement network segmentation, access controls, and cybersecurity incident response plans. These directives address the IT-OT boundary vulnerability but do not eliminate the voluntary shutdown risk: an operator whose SCADA network shows signs of sophisticated pre-positioning — lateral movement, anomalous historian access, command injection attempts — will shut down before the adversary executes, precisely because the Colonial precedent established the legal and reputational consequences of continuing to operate under suspected compromise.

2.4 Major Pipeline Systems — Consequence Profile

System	Type	Capacity (bpd approx.)	Consequence if Lost
Colonial Pipeline (Lines 1 & 2)	Product	~2,500,000	Eastern Seaboard gasoline, diesel, jet fuel — 45% of regional supply. Shortages in Southeast within 5 days, Mid-Atlantic within 10 days, New England within 14 days.
Enbridge Mainline (Lines 1–6)	Crude	~3,000,000	Primary Canadian and Bakken crude supply to Midwest and Gulf Coast refineries. Refinery feedstock curtailment within days; no alternative crude routing at equivalent volume.
Dakota Access Pipeline	Crude	~750,000	Bakken crude supply to Gulf Coast markets. Loss forces Bakken producers to rail — higher cost, lower volume, longer transit.
Plantation Pipeline	Product	~600,000	Southeast product supply parallel to Colonial. Partial redundancy with Colonial but cannot absorb full Colonial volume loss.
Seaway Pipeline	Crude	~850,000	Cushing, Oklahoma to Gulf Coast refineries. Loss isolates Cushing crude storage from Gulf Coast refinery demand.

2.5 Regulatory Authority

PHMSA. The Pipeline and Hazardous Materials Safety Administration (49 CFR Part 195) regulates liquid pipeline safety: design standards, pressure testing, integrity management, control room management, and emergency response planning. PHMSA's integrity management program requires operators to assess and mitigate risks at high-consequence areas — locations where a pipeline failure could affect populated areas, drinking water sources, or environmentally sensitive areas. PHMSA does not regulate pipeline security; its jurisdiction is safety, not protection from deliberate attack.

TSA. Following the Colonial Pipeline ransomware attack, TSA issued a series of mandatory Security Directives for pipeline operators covering cybersecurity incident reporting, network segmentation, access controls, and cybersecurity incident response planning. These directives represent the first mandatory cybersecurity framework for petroleum pipeline operators. They do not address physical security; physical security requirements for pipeline right-of-way, pump stations, and control centers remain voluntary. The TSA directives provide the regulatory foundation on which the physical security design basis for FC4 must be built.

EPA SPCC and OPA 90. Environmental regulations under the Oil Pollution Act of 1990 and the SPCC rule require pipeline operators to prepare facility response plans for worst-case discharge scenarios and to maintain oil spill response capability. For pipeline right-of-way ruptures at environmentally sensitive crossings, the OPA 90 response framework governs the remediation process and timeline. PHMSA restart authorization following a significant liquid release requires hydrostatic pressure testing, which adds weeks to the physical repair timeline independent of the time required to complete the repair itself.

3. Design Basis: Crude and Product Transport Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the pipeline facility class inherits the infrastructure failure event, operational event, and natural hazard event categories from the DBA-OIL Sector Framework. This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

Pipeline rupture at a high-consequence crossing. The governing infrastructure failure bounding event is a large-volume release from a major liquid pipeline at a high-consequence area crossing — a navigable waterway, a drinking water source, or a populated area. The 2010 Enbridge Line 6B rupture in Marshall, Michigan is the reference event: a corroded pipeline segment failed at a wetland crossing, releasing approximately 843,000 gallons of diluted bitumen into the Kalamazoo River. The cleanup required four years and approximately one billion dollars. The PHMSA response framework — pipeline isolation, environmental assessment, hydrostatic pressure testing, and regulatory restart authorization — produces a restart timeline that significantly exceeds the physical repair duration. At major watercourse crossings where

environmental remediation requirements are most stringent, the regulatory restart timeline can extend six to twelve months beyond the physical repair completion.

SCADA control system failure. Loss of the pipeline SCADA system governing a major trunk line, requiring emergency shutdown and transition to manual operations. Unlike a refinery DCS failure, which affects a single facility, a trunk pipeline SCADA failure affects the entire pipeline simultaneously — all pump stations lose coordinated control, all batch tracking is lost, and product custody transfer cannot be verified. Manual operations on a major product pipeline are not feasible at normal throughput; the pipeline must be shut down until SCADA integrity is restored. The bounding case involves a SCADA failure during a batch interface period, producing product commingling that requires extensive product quality testing and interface management before any restart is possible.

Pump station fire. A major fire at a pump station driven by a seal failure or piping leak at a high-pressure pump, resulting in destruction of the pump unit and associated electrical infrastructure. The pump station fire scenario is the non-adversarial analog of the kinetic pump station attack in the DBA-MA extension. The repair timeline — pump unit procurement, installation, electrical infrastructure reconstruction, and PHMSA inspection — governs the pipeline restart at the affected segment, typically 8 to 20 weeks depending on the pump unit specification and supply chain status.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the crude and product transport facility class is a coordinated, multi-vector attack against a major pipeline system or intersecting pipeline systems, combining SCADA compromise with precision kinetic action against pump stations at hydraulic chokepoints, timed to exploit a global supply disruption context that degrades the import and emergency response mechanisms available to limit the domestic consequence.

The bounding parameters, inherited from the sector framework threat taxonomy:

Adversary tier: Tier 1 state actor with advanced cyber pre-positioning capability and precision kinetic delivery. The voluntary shutdown dynamic means that a sophisticated Tier 2 actor with SCADA access alone can produce a consequence-equivalent event without kinetic action; the DBA-MA bounding event adds kinetic action to extend the outage beyond SCADA reconstitution.

Modality: Cyber-kinetic combined arms. SCADA compromise — achieved through multi-stage intrusion over preceding months — initiates the voluntary shutdown dynamic or directly manipulates pump station controls to create over-pressure conditions. Kinetic action against pump stations at hydraulic chokepoints executes simultaneously, ensuring that physical damage prevents restart even after SCADA reconstitution is complete. The cyber component maximizes the disruption onset; the kinetic component maximizes the disruption duration.

Target selection: Pump stations at hydraulic chokepoints — terrain-constrained locations where pipeline elevation change, major river crossings, or single-path corridor constraints mean that loss of that station stops all flow through the affected segment. For Colonial Pipeline, the Appalachian crossing stations represent the governing chokepoints: there is no alternative routing around the mountain terrain, and loss of pumping capacity at those stations stops product flow to the entire Mid-Atlantic and Northeast market.

Timing: Coordinated with a Hormuz closure or Persian Gulf supply disruption that has already tightened the global crude market, elevated domestic crude prices, and constrained the import alternatives that would normally allow the domestic supply chain to absorb a pipeline outage. Under these conditions, the SPR drawdown that would be the primary federal response to a domestic pipeline outage may already be committed to offsetting the international supply disruption, reducing its availability as a domestic response tool. The adversary who initiates both simultaneously compounds the consequence of each.

Simultaneity: Coordinated attacks on at least two pipeline systems serving the same market — a crude trunk line and a product pipeline — within a 48-hour window. The crude trunk line attack constrains refinery feedstock; the product pipeline attack simultaneously constrains product delivery to the market those refineries serve. The compound consequence eliminates the supply chain redundancy that each system provides for the other.

3.3 Dual Design Basis Architecture

Consistent with the DBA-OIL Sector Framework (§5.4) and DBA-EN-SF (§4.2), a crude and product pipeline system may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

Three pipeline-specific AI governance interaction points are identified for the FC4 Design Basis:

Pipeline SCADA and predictive control systems. SCADA systems that manage pump station operations, pressure profiles, and batch tracking across a trunk pipeline are ML systems operating below the Bright Line when performing deterministic pressure management and flow control functions within their trained operating envelopes. They are formally testable and may operate autonomously. However, an adversarially manipulated SCADA output that drives pipeline operating pressure above the maximum allowable operating pressure (MAOP) at a targeted segment can initiate a rupture — the physical damage that the cyber component of the DBA-MA bounding event is designed to create. The AI Governance Design Basis must establish that SCADA pressure management outputs are range-checked against independent pressure sensor data before pump commands are executed, that anomalous pressure trajectories trigger automatic pump shutdowns independent of SCADA command confirmation, and that the

SCADA system's operating envelope validation is performed by a security-reviewed process that is architecturally isolated from the corporate IT network.

Anomaly detection and leak detection ML. Computational pipeline monitoring (CPM) systems use ML algorithms to detect pipeline leaks by identifying statistical anomalies in pressure, flow rate, and product balance data across the pipeline network. These systems are ML operating below the Bright Line when performing signal classification within their trained envelopes. The adversarial concern is the manipulation of CPM inputs — through false sensor data injected via compromised SCADA — to mask a deliberate rupture or to generate false positive leak signals that trigger unnecessary shutdowns. The Design Basis requirement is that CPM systems receive sensor data through paths that are architecturally independent of the SCADA command path, that anomalous sensor patterns inconsistent with normal pipeline operations trigger investigation before shutdown rather than automatic shutdown, and that CPM model integrity is validated against a defined adversarial input scenario set at each model update cycle.

Pipeline scheduling and optimization, Gen AI. Pipeline scheduling systems that optimize product batch sequencing, pump station dispatch, and delivery nominations increasingly incorporate Gen AI advisory tools. These tools operate above the Bright Line: they are non-deterministic and not formally testable. A human Command Broker must be in the control loop for any pipeline operating parameter change recommended by a Gen AI scheduling tool. In an adversarial context, a compromised Gen AI scheduling tool that recommends batch sequences designed to create pressure transients at targeted pipeline segments is a cyber attack pathway that operates through the pipeline's own optimization infrastructure. The Design Basis requires that pipeline operators maintain a manual scheduling capability that does not depend on Gen AI advisory tools, and that the Command Broker function for Gen AI scheduling recommendations is performed by a qualified pipeline controller whose authorization is local, auditable, and not itself executable by a software component.

4. Design Basis Accident Scenarios

4.1 Scenario A: Cyber-Kinetic Attack on a Major Product Pipeline

4.1.1 Initiating Event

Coordinated cyber-kinetic attack against Colonial Pipeline, targeting the Appalachian crossing pump stations that represent the hydraulic chokepoint between Gulf Coast supply and Mid-Atlantic and Northeast markets. The cyber component, having achieved access to Colonial's pipeline control network through a multi-stage intrusion campaign that exploited the IT-OT boundary over the preceding months, initiates anomalous pressure readings at multiple pump stations simultaneously — mimicking the signature of a pipeline rupture to trigger precautionary shutdown protocols. The SCADA manipulation is designed to force a voluntary operator

shutdown before any kinetic action occurs, ensuring the pipeline is offline, and the control room is in emergency response mode when the physical attack executes.

The kinetic component executes within 30 minutes of the SCADA-induced shutdown: precision UAS strikes against three pump stations at the Appalachian crossing — the Sherwood, Roanoke, and Newington stations, or equivalent chokepoint stations — destroying the primary pump units, electrical switchgear, and local control infrastructure at each. The physical destruction is calibrated to prevent restart even after SCADA forensic investigation and reconstitution are complete: the pump units must be procured, shipped, and installed before any product flow can resume through the mountain corridor.

4.1.2 Progression Sequence

T+0: SCADA anomaly triggers precautionary pipeline shutdown — the voluntary shutdown dynamic executes as designed. The control room in Alpharetta initiates emergency shutdown procedures. TSA and CISA notification initiated. All pump stations are secured remotely. Product batches in transit halt in place.

T+0 to T+30 minutes: Kinetic strikes against three Appalachian crossing pump stations. Physical destruction confirmed by remote sensing and station communications loss. The control room simultaneously manages the SCADA anomaly investigation and the physical attack response. DOE ESF-12 notification. FBI and DOD notification for confirmed state-actor kinetic attack on critical infrastructure.

T+30 minutes to T+72 hours: SCADA forensic investigation initiated — minimum 4 to 6 weeks before full SCADA integrity can be confirmed. Physical damage assessment at the three pump stations is underway. Pump unit specifications and procurement inquiries initiated. Emergency product supply assessment for Eastern Seaboard markets: Plantation Pipeline activated to maximum capacity; marine product import authorization requested; SPR crude drawdown assessed for Gulf Coast refinery throughput maintenance. Southeast retail fuel shortages are appearing within 72 hours.

T+72 hours to 8 weeks: SCADA forensic investigation continuing. Pump unit procurement underway — custom pump units for Colonial's Appalachian stations have a lead time of 14 to 20 weeks under normal conditions. Emergency marine product imports are increasing in the Northeast markets. Mid-Atlantic shortages are developing. DOD emergency fuel allocation activated for military installations along the Eastern Seaboard. A Jones Act waiver is requested for foreign-flagged product tankers to supplement the domestic marine supply.

T+8 weeks to recovery: SCADA reconstitution completed; pipeline cannot restart until pump station physical repairs are complete. Pump unit installation is underway at the three Appalachian stations sequentially — contractor and equipment availability limits parallel installation. PHMSA inspection and restart authorization adds 2 to 4 weeks to the physical installation timeline. Full Colonial Pipeline restart 5 to 8 months after the attack.

4.1.3 Consequence Envelope

Supply disruption duration. The governing recovery constraint is the pump unit procurement and installation timeline at the Appalachian chokepoint stations, not the SCADA reconstitution timeline. SCADA reconstitution is complete in 4 to 6 weeks; the pump units require 14 to 20 weeks from procurement to installation. The compound timeline — procurement, delivery, installation, and PHMSA inspection — produces a full Colonial Pipeline outage of 5 to 8 months. This is qualitatively different from the 6-day Colonial ransomware event: it is a supply disruption of a duration and scale that the available response mechanisms — Plantation Pipeline backup, marine imports, SPR drawdown — can partially but not fully offset.

Market consequence. Eastern Seaboard retail fuel prices spike within 72 hours of the attack's public confirmation. Southeast markets experience shortfalls first; Mid-Atlantic and Northeast markets follow within two weeks. Aviation fuel supply at military and commercial airports along the Eastern Seaboard is prioritized over civilian motor fuel supply, accelerating the civilian shortfall. The economic consequence of a 5 to 8 month Colonial Pipeline outage — measured in fuel price increases, GDP impact, and military logistics disruption — substantially exceeds any prior domestic energy infrastructure disruption.

Global supply context amplification. If the attack is timed to coincide with a Hormuz closure or Persian Gulf supply disruption, the consequence envelope expands in two dimensions. First, the SPR drawdown that would normally offset a domestic crude supply disruption may already be partially committed to the international response, reducing available volumes for domestic refinery feedstock support. Second, the marine product import surge that represents the primary Eastern Seaboard supply bridge draws on global product tanker availability already strained by the Persian Gulf disruption. The adversary who understands this compound dynamic uses the international supply stress as a force multiplier for the domestic pipeline attack, and the timing is available to any state actor with sufficient intelligence on both the domestic pipeline control system and the international supply chain.

4.2 Scenario B: Multi-Point Kinetic Rupture of a Major Crude Trunk Pipeline

4.2.1 Initiating Event

Coordinated kinetic attacks against a major crude trunk pipeline — the Enbridge Mainline or Dakota Access Pipeline — at four to six locations chosen to exceed the pipeline operator's simultaneous repair capacity and to include at least two environmentally sensitive crossings where PHMSA's regulatory restart requirements impose a timeline significantly longer than the physical repair. The attack employs shaped charges at buried pipeline segments at watercourse crossings, timed to execute within a six-hour window. Each rupture produces a crude oil release to the surrounding environment; the releases at the watercourse crossings reach navigable waterways.

4.2.2 The Repair Capacity Constraint

A major crude pipeline operator can typically mobilize pipeline repair crews and equipment to address one or two simultaneous ruptures anywhere along the right-of-way within 24 to 48 hours. Four to six simultaneous ruptures exceed this capacity: crews and equipment must be sequenced across the repair sites, and the specialized equipment required for watercourse crossing repairs — bypass pumping, environmental containment, diving, or directional drilling for beneath-waterway repairs — is available in finite quantities. The adversary who designs the six-rupture attack understands that the sixth rupture's repair timeline is determined not by the time required to repair that rupture in isolation, but by when the repair crew and equipment queue reaches it.

At the watercourse crossing ruptures, the repair timeline extends further: PHMSA requires hydrostatic pressure testing and regulatory inspection before restart at high-consequence area crossings. For a major navigable waterway crossing where environmental remediation is required in addition to physical repair, the combined physical-plus-regulatory timeline can reach six to twelve months per location, regardless of how quickly the physical repair is executed.

4.2.3 Consequence Envelope: Crude Trunk Rupture

Refinery feedstock impact. A major crude trunk pipeline carrying 500,000 to 1,000,000 barrels per day offline for six to twelve months removes a substantial fraction of domestic crude pipeline capacity. Refineries dependent on that pipeline for feedstock must reduce throughput, switch to alternative crude grades requiring configuration adjustments, or draw down crude storage — none of which fully compensates at equivalent volume and timeline. The compound effect on refined product output begins within two to three weeks of the pipeline shutdown as crude storage buffers are exhausted.

Environmental consequence cascade. Multiple simultaneous crude oil releases to navigable waterways trigger parallel USCG Federal On-Scene Coordinator activations, EPA response, and state environmental response across multiple jurisdictions simultaneously. The federal environmental response system is not sized for four simultaneous major pipeline spills. Response resource allocation — boom, skimmers, cleanup contractors — requires federal coordination to prioritize the highest-consequence releases while managing the others. The environmental remediation timeline at major watercourse crossings is measured in years, not months, and represents an ongoing legal and reputational consequence long after pipeline operations have resumed.

Allied crude supply consequence. Major crude trunk pipelines carry not only domestic production but also Canadian crude imports — the Enbridge Mainline is the primary conduit for Canadian oil sands production entering the U.S. market. A successful attack on the Enbridge Mainline disrupts Canadian crude supply to U.S. refineries, producing diplomatic and trade consequences in addition to the supply disruption. Canada, as a NATO ally and the largest

supplier of U.S. crude imports, has a direct energy security interest in the protection of the transboundary pipeline infrastructure that connects the two supply systems.

4.3 Scenario C: Compound Attack on Intersecting Pipeline Corridors

4.3.1 Initiating Event

Coordinated attacks against both a major crude trunk pipeline and a major product pipeline sharing a common corridor or serving the same regional market, executed within a 48-hour window. The scenario employs Scenario A's cyber-kinetic modality against the product pipeline and Scenario B's multi-point kinetic modality against the crude trunk line simultaneously. The defining characteristic is the elimination of the supply chain redundancy that each system normally provides for the other. When both are offline, the downstream market has neither feedstock supply nor product delivery capability, and the upstream refinery system has neither the crude to process nor the pipeline capacity to deliver what it produces.

4.3.2 The Compound Supply Chain Failure

In normal operations, the crude trunk pipeline and the product pipeline serve complementary roles in a single supply chain: the crude line feeds the refinery, and the product line delivers what the refinery produces. Each provides partial insurance against the other's failure — a crude trunk disruption can be partially offset by drawing down crude storage while alternative routing is arranged; a product pipeline disruption can be partially offset by building product inventory at origin terminals while marine alternative supply is arranged. These mitigation mechanisms depend on the other system being operational.

A simultaneous attack on both systems eliminates the mitigation capacity of each. Crude storage drawdown during the trunk line outage cannot be sustained if the refinery is also curtailing throughput because it cannot deliver product. Product inventory built at origin terminals cannot offset the product pipeline outage if crude supply to those refineries is simultaneously constrained. The compound failure produces a supply consequence that is not the sum of two independent outages but the product of their interaction — a multiplicative consequence that the single-system emergency response framework is not designed to address.

4.3.3 The Hormuz Closure Compound

The Scenario C compound consequence is the scenario where the Hormuz closure or Persian Gulf supply disruption context produces its most material effect. Under normal global supply conditions, a simultaneous crude-and-product pipeline outage in the Gulf Coast corridor produces a severe but manageable supply disruption — the global market provides import alternatives for both crude and products, and the SPR provides a crude bridge during the repair period.

Under a Hormuz closure or sustained Persian Gulf supply disruption, this global supply buffer is degraded across three dimensions. Import alternatives for both crude and refined products are constrained by the global market tightness produced by the Persian Gulf disruption. The SPR, which may already be drawing down to offset the international crude supply reduction, has reduced availability for domestic pipeline disruption response. And the political and economic environment — elevated energy prices, allied supply stress, military logistics pressure — amplifies every unit of domestic supply disruption into a larger strategic consequence than it would produce under normal market conditions.

The adversary who designs the Scenario C attack to coincide with a Hormuz closure is not simply attacking two pipelines simultaneously. It is exploiting the compound vulnerability of a domestic supply chain that has lost its international buffers at precisely the moment its most critical domestic transport infrastructure is destroyed. The Design Basis must account for this compound scenario as the bounding consequence event, not as a low-probability edge case.

4.3.4 Consequence Envelope: Compound Attack

Supply impact. Simultaneous loss of a major crude trunk line (500,000 to 1,000,000 barrels per day) and a major product pipeline (Colonial equivalent: 2,500,000 barrels per day equivalent in product volume) during a Hormuz closure / Persian Gulf supply disruption. The compound domestic supply disruption — refinery feedstock constrained by the crude line outage, product delivery constrained by the product line outage, and import alternatives constrained by global market tightness — produces a consumer fuel supply crisis without precedent in U.S. peacetime energy history. Mandatory demand allocation, military fuel priority rationing, and emergency import authorization are all activated simultaneously, competing for federal decision-making attention and response resources.

Recovery timeline. The governing constraint is the product pipeline pump station replacement (5 to 8 months per Scenario A) and the crude trunk pipeline watercourse crossing repair and regulatory authorization (6 to 12 months per Scenario B), running in parallel but with no mutual offset. The compound outage duration is determined by whichever recovery timeline is longer and by the availability of the specialty contractors and replacement equipment that both recovery programs compete for simultaneously. Under the Hormuz closure context, global supply tightness persists throughout the domestic pipeline recovery period, providing no market relief as the physical infrastructure is rebuilt.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in U.S. pipeline SCADA systems or specific pipeline operator control architectures. Anomalous pipeline control network activity: unusual SCADA historian queries, anomalous vendor access sessions outside maintenance windows, and lateral movement between IT and OT network segments inconsistent with normal operations. Physical surveillance of pump station facilities or right-of-way access points at known hydraulic chokepoints. Procurement inquiries for pump unit specifications or pipeline engineering drawings for specific pipeline systems. Open-source reporting of adversary pre-positioning activity against U.S. critical infrastructure control systems coinciding with international supply disruption events.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized access to pipeline SCADA or control system networks. Simultaneous anomalous pressure or flow readings at multiple stations on the same pipeline system are inconsistent with operational variation. Confirmed physical intrusion at a pump station facility. Simultaneous security incidents at pump stations on two or more pipeline systems within 48 hours. Confirmed kinetic event at any pump station or pipeline right-of-way. SCADA anomaly patterns consistent with the Colonial Pipeline intrusion signature — historian access, lateral movement, command staging — at any major pipeline operator.

5.2 Decision Authority Framework

Pipeline operator level: Pipeline controller authority to initiate emergency shutdown of individual pump stations or entire pipeline segments based on SCADA anomaly or physical incident. TSA notification within 12 hours of confirmed or suspected cybersecurity incident per TSA Security Directive requirements. PHMSA notification for pipeline releases. USCG and EPA notification for releases reaching navigable waters. FBI notification for confirmed physical attacks or suspected state-actor cyber intrusion.

Federal coordination: CISA coordinates federal cyber incident response for pipeline SCADA intrusions. DOE ESF-12 activation for supply disruption response. EPA and USCG Federal On-Scene Coordinator for environmental releases. FBI and DOD for events with confirmed or strongly suspected state-actor attribution. NSC coordination for Scenario C compound events involving simultaneous attack on multiple pipeline systems.

International coordination: For attacks on transboundary pipeline systems — Enbridge Mainline and other Canada-U.S. pipeline crossings — immediate notification to Canadian counterparts through the Canada-U.S. Critical Infrastructure Protection framework. For compound events involving Hormuz closure, the State Department and allied energy security coordination through the International Energy Agency emergency response mechanism.

5.3 Federal Coordination Thresholds

Confirmed kinetic attack on any pump station or pipeline segment of a major crude trunk or product pipeline system. Confirmed SCADA intrusion with suspected nation-state attribution at any major pipeline operator. Simultaneous anomalous SCADA events at pump stations on two or more major pipeline systems within 48 hours. Pipeline release to a navigable waterway above the PHMSA significant incident reporting threshold from a cause consistent with adversary action. Product supply disruption affecting Colonial Pipeline or an equivalent major product system for more than 72 hours. A compound event involving simultaneous crude trunk and product pipeline disruptions affecting the same regional market.

6. Recovery Architecture

6.1 Pipeline Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency pipeline shutdown and pressure relief. Environmental release containment and agency notification. SCADA forensic investigation initiation — this is a Phase 1 action, not Phase 2, because the SCADA reconstitution timeline is the critical path for cyber-component events. Physical damage assessment at the attacked pump stations. Emergency product supply routing activation: Plantation Pipeline to maximum capacity, marine import authorization, and DOE ESF-12 activation. Pump unit specification and procurement inquiry initiated — lead time assessment is a Phase 1 action because the procurement timeline gates the physical recovery.

Phase 2 — Short-term (72 hours to 8 weeks). SCADA forensic investigation continuing. Physical damage assessment complete. Pump unit procurement is underway. Environmental remediation was initiated at the release sites. Alternative supply arrangements formalized — marine import volumes, emergency DOD fuel logistics activated. PHMSA coordination for restart authorization planning, including identification of which segments require hydrostatic testing and which can be restarted without full testing under emergency provisions.

Phase 3 — Medium-term (8 weeks to recovery). SCADA reconstitution completed. Pump unit installation at affected stations. PHMSA inspection and restart authorization for each repaired segment. Sequential pipeline segment restart as each station repair and authorization is completed. Supply chain monitoring for adversary interdiction of pump unit components — the adversary who understands the recovery critical path may attempt to extend the outage by interdicting pump unit procurement.

6.2 Critical Recovery Constraints

Pump unit procurement and installation. Custom pump units for major pipeline systems are engineered to specific flow rates, pressures, and driver configurations. Lead times of 14 to 20 weeks under normal supply chain conditions reflect both manufacturing time and the limited number of manufacturers producing large centrifugal pumps for high-pressure liquid pipeline

service. A pre-positioned strategic spare pump program — analogous in concept to the electric sector's spare transformer reserve — would reduce this constraint materially. No such program currently exists for liquid pipeline pump infrastructure.

SCADA forensic investigation and reconstitution. A minimum of 4 to 6 weeks for major pipeline SCADA forensic investigation and reconstitution under the TSA Security Directive incident response framework. This timeline cannot be compressed without compromising the forensic integrity required for attribution and legal proceedings, and for the operator's own assurance that the intrusion has been fully remediated before resuming automated operations. The SCADA reconstitution timeline runs in parallel with physical repairs for Scenario A events, but is the sole governing constraint if the kinetic component is limited or absent.

PHMSA restart authorization. Following any significant pipeline incident — particularly releases at high-consequence area crossings — PHMSA requires operator notification, incident investigation, integrity assessment, and, in many cases, hydrostatic pressure testing before restart authorization. The regulatory timeline adds 2 to 6 weeks to the physical repair completion date and cannot be waived without legislative action. Emergency provisions exist for pressure reduction and partial service restoration under PHMSA's operating allowance framework, but full pressure restoration requires completed regulatory review.

Environmental remediation at watercourse crossings. Crude oil releases at navigable waterway crossings initiate environmental remediation processes that extend for years beyond the pipeline restart. The remediation timeline does not gate pipeline restart — the pipeline can be repaired and restarted while remediation continues — but the cost, legal liability, and regulatory attention associated with ongoing remediation affect the operator's ability to focus resources on other recovery activities and create political pressure that complicates the restart authorization process.

6.3 Global Supply Context and Recovery

For Scenario C compound events timed to coincide with a Hormuz closure or Persian Gulf supply disruption, the recovery architecture must account for a global supply environment that is simultaneously degraded. The International Energy Agency emergency response mechanism — coordinated stock drawdown across IEA member states — is the primary global supply response tool; its activation during a Persian Gulf disruption reduces the available IEA stock drawdown capacity for domestic U.S. pipeline recovery support. Recovery planning for compound Scenario C events must explicitly account for the IEA mechanism's reduced availability and identify alternative supply bridge mechanisms — expanded domestic production, demand management, and allied bilateral supply agreements — that do not depend on IEA stock drawdown.

7. Gaps and Recommendations

7.1 Regulatory Gap: Absence of Mandatory Physical Security Standards for Pipeline Infrastructure

TSA's mandatory cybersecurity directives for pipeline operators, issued following the Colonial Pipeline ransomware attack, represent a significant advance in pipeline security regulation. They do not address physical security. A pipeline operator whose pump stations are the hydraulic chokepoints for regional fuel supply has no regulatory requirement to assess their vulnerability to UAS attack, precision kinetic action, or vehicle-borne explosive. TSA should develop mandatory physical security standards for pipeline pump stations and control centers, using the pump station vulnerability assessment framework developed for the Colonial Pipeline event as the regulatory model. The cybersecurity directive framework provides the legislative authority and enforcement mechanism; extending it to physical security requires a regulatory determination that physical attack is within TSA's pipeline security jurisdiction, which the Colonial Pipeline kinetic threat environment clearly supports.

7.2 Pump Unit Strategic Spare Program

The 14 to 20 week pump unit procurement timeline is the governing physical recovery constraint for Scenario A and the compound Scenario C event. No federal or industry program pre-positions critical spare pump units for major pipeline systems. The electric sector's STEP program (Spare Transformer Equipment Program) provides the analytical model: a federally supported strategic spare equipment program for the longest-lead recovery-critical components. DOE, in coordination with TSA and PHMSA, should assess the feasibility of an analogous program for major pipeline pump units, prioritizing the hydraulic chokepoint stations on systems whose loss produces the most consequential supply disruptions. The Colonial Pipeline Appalachian crossing stations and the Enbridge Mainline river crossing stations are the highest-priority candidates for pre-positioned spare equipment.

7.3 Compound Event Response Framework

No federal emergency response framework addresses the compound consequence of simultaneous crude trunk and product pipeline disruptions affecting the same regional market during a global supply disruption event. DOE ESF-12, TSA pipeline security response, PHMSA incident response, and State Department / IEA international energy coordination are managed by separate agencies with no common scenario planning baseline for Scenario C. DOE should lead the development of a compound petroleum supply disruption response framework that integrates domestic pipeline recovery, international energy security coordination, and military fuel priority allocation into a single planning document with defined decision authorities and activation thresholds.

7.4 Transboundary Pipeline Security Coordination

The Enbridge Mainline and other Canada-U.S. transboundary crude pipeline systems carry supply that is strategically important to both nations. A successful attack on transboundary pipeline infrastructure affects Canadian crude export revenues and U.S. refinery feedstock supply simultaneously, creating a shared security interest that is not reflected in any bilateral security agreement specifically addressing pipeline infrastructure. The existing Canada-U.S. Critical Infrastructure Protection framework provides the diplomatic foundation; a bilateral pipeline security annex — covering threat information sharing, joint vulnerability assessment, and coordinated emergency response — should be developed before a compound event creates the reactive conditions for poorly designed bilateral arrangements.

7.5 Hormuz Closure Compound Scenario Planning

No domestic energy emergency response plan explicitly addresses the compound consequence of a Hormuz closure or Persian Gulf supply disruption coinciding with a major domestic pipeline attack. The two response mechanisms — IEA emergency stock drawdown for the international disruption, SPR drawdown and pipeline emergency response for the domestic disruption — are managed independently and have not been stress-tested against a scenario that activates both simultaneously. DOE, in coordination with the NSC and the IEA, should develop a compound scenario planning baseline that establishes SPR drawdown priorities, IEA coordination protocols, and domestic demand management triggers for a scenario in which both the international and domestic response mechanisms are activated simultaneously. The Scenario C consequence envelope — with the Hormuz compound timing — provides the appropriate design basis for that planning effort.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the crude and product transport facility class. Issued under DBA-OIL Sector Framework v1.5. Three scenarios developed: Scenario A (cyber-kinetic attack on a major product pipeline — Colonial template with kinetic amplification at Appalachian chokepoint pump stations), Scenario B (multi-point kinetic rupture of a major crude trunk pipeline — exceeding simultaneous repair capacity with environmentally sensitive crossing selection), Scenario C (compound attack on intersecting pipeline corridors with Hormuz closure amplification). Dual Design Basis architecture established with three pipeline-specific AI governance interaction points. Regulatory gaps identified: mandatory physical security standards for pipeline infrastructure, pump unit strategic spare program, compound event response framework, transboundary pipeline security coordination, and Hormuz closure compound scenario planning gap. Completes the initial DBA-OIL facility class instrument set: FC1 through FC4 all at v0.1 or above.

— *END OF DOCUMENT* —